Introduction to Software Engineering

COSC 336

ReSource (Phases IV and V)

Design Document

Architecture Design and Detailed Design

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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Sections in This File

This file contains the parts of the ReSource Phase 4 and 5 Design Document prepared by Ahmed AlAbdouli (100062836). Section, figure and table numbers are the same as in the combined group document.

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4. AI Services

Prepared by: Ahmed AlAbdouli

ReSource has six AI services: the five named in the project description (classification, semantic matching, ranking, recommendation engine and LLM assistant) and the AI report generator required by FR-12. Table 4-1 gives, for each service, the call pattern, why we chose it, the time limit, what happens when the service is down and what happens when it is unsure. The patterns are those of the Phase 4 guide [2]: A, ask and wait; B, compute ahead; C, queue and notify.

Table 4-1: AI services, call patterns and fallbacks

Service	Pattern	Why	Time limit	If down	If unsure (where Low starts)
ClassificationService (FR-9, UC-10): category, tags, description, missing fields	A	The Asset Coordinator is on the form and can always choose the category from the list without it.	Progress shown within 1 s; result within 15 s for 90% (PER-4); cut off at 20 s (PER-5)	Fields stay empty, entries are kept; the coordinator picks from the approved list (FR-9.8)	Calibrated confidence below 0.50, or the local vote and the language model disagree: shown as Low, nothing pre-filled; never pre-accepted at any level (FR-9.6)
EmbeddingService and MatchingEngine (FR-8.1): text vectors for assets and requests	B	Each asset is embedded once when saved and read by every later request; the request is embedded on save.	Search over vectors under 1 s; embedding job retried 3 times (after 1, 5 and 15 min)	Keyword and rule search, results labeled "not ranked"; the asset waits for the retry	Cosine similarity below 0.30 gives a function score of 0, so the asset cannot reach the top of the list
RankingModel (FR-8.3 to FR-8.6, UC-9)	A	The Procurement Team member (or the requester on the request form) is waiting for the list.	Ranking itself under 2 s; whole match page within 15 s for 90%, cut off at 20 s	Keyword and rule matches only, labeled "not ranked", no scores (AI-6)	No candidate scores 60 or more: "weak matches" warning and no top pick; each match also has its own confidence level (Low below 0.50)
ActionRecommender (FR-10, Desirable, UC-7)	C	Nobody waits: it runs when an inspection starts, a repair is closed as not repairable, the cost flag is raised (FR-5.11) or an end-of-life proposal is opened; the suggestion is ready when a person reviews it.	Suggestion ready within 5 minutes of the trigger; 3 retries	No suggestion; the proposal shows the asset summary for manual review (FR-10.8)	Low (below 0.50) is marked; for recycling, retirement or disposal the approver must record their own reason (SAF-6); never applied by itself
AssistantService (FR-11, Desirable, UC-11)	A	The Department Representative waits in the chat.	Progress within 1 s; first answer within 15 s for 90%; cut off at 20 s	"The assistant is unavailable"; the draft is kept and opens in the request form; search still works (FR-11.9)	Answers only from permitted records returned by its read tools; if none support an answer it says so (QUA-15); if the request is unclear (below 0.50) it asks a question instead of drafting; every action needs a confirm (FR-11.3)
ReportGenerator (FR-12, UC-12)	A	The officer waits for the draft; PER-4 includes AI reports in the 15-second target.	Progress within 1 s; draft within 15 s for 90%; cut off at 20 s	Message, then the standard report and indicators for the same period open (FR-12.11)	Any figure that differs from the standard figures rejects the draft (FR-12.4); each recommended action shows its confidence and Low ones are marked

4.1 AIGateway

Prepared by: Ahmed AlAbdouli

Every AI call goes through AIGateway (Rule 3). The gateway does the same six things for every service, so no module has to repeat them:

1. **Checks access first.** It receives the user with each call and passes on only records that user may see (AI-7).
2. **Removes personal data.** It builds the payload from the field allow-list in Section 16.2 and removes e-mail addresses, university ID patterns and names of registered users from free text (SEC-11).
3. **Applies the time limits.** It shows progress within 1 second and stops waiting at 20 seconds (PER-5). After three failures of the same service within five minutes it marks the service as down for two minutes and answers "unavailable" at once, so users do not wait 20 seconds each time.
4. **Sets the confidence level.** It turns the calibrated confidence into High, Medium or Low with the versioned thresholds in Table 4-2 (AI-10).
5. **Treats text as data.** Instructions found inside asset descriptions, messages or files cannot change permissions or approval rules (AI-9); the services can only read through permission-filtered tools and can never write a business record.
6. **Records every output.** It saves the feature, input record references, model and rule versions, result, confidence, explanation and time taken, and later the human decision (AI-5, FR-7.A4).

Table 4-2: Confidence levels (rule set v3)

Level	Calibrated confidence	Shown to the user as	Effect
High	0.80 or more	Text "High" and the number	Normal suggestion; still needs a person to accept
Medium	0.50 to below 0.80	Text "Medium" and the number	Normal suggestion with its main limitation
Low	Below 0.50, or required information missing or conflicting	Text "Low", the number and the reason	Not pre-filled; disposal-type advice needs the approver's own reason (SAF-6)

Confidence is always shown as text as well as color (QUA-3). The thresholds are those of AI-10 and are stored with the model version (Section 19); they are confirmed against evaluation data before release (TBD-12).

4.2 How Each Service Works

Prepared by: Ahmed AlAbdouli

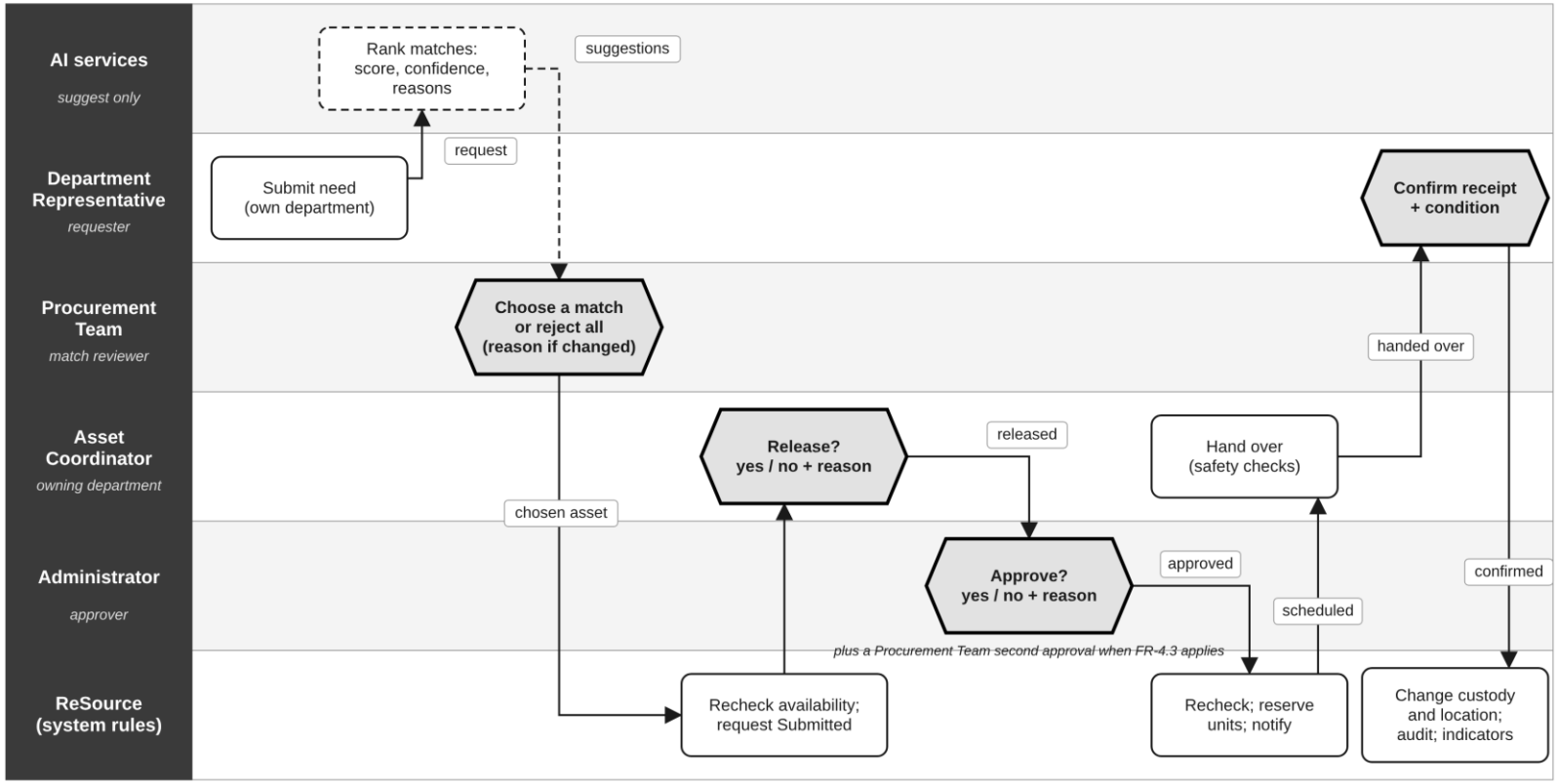
- **ClassificationService.** The local embedding model finds the five most similar already-classified assets and votes for a category; the language model suggests a category from the approved list, tags, a factual description and the missing fields. If both agree, the confidence is the calibrated vote; if they disagree, the level is Low and both options are shown with the reason (FR-9.5). With fewer than five labeled neighbours the level is at most Medium.

- **EmbeddingService and MatchingEngine.** The worker embeds name, description, specifications, category and tags with all-MiniLM-L6-v2 [10], [11], a 384-dimension sentence model released under the Apache 2.0 license, and stores the vector with a hash of the text, so an unchanged asset is not embedded again.
- **RankingModel.** A documented, deterministic score of eight factors (Section 17) ranks the 20 most similar eligible assets. The weights are visible to Administrators and do not include the department name or size (LEG-9).
- **ActionRecommender.** It scores the eight actions of FR-10.1 with rules on condition, safety, maintenance cost, demand (open requests in the same category) and remaining life, prefers keeping assets in use (FR-10.3) and shows cost, weight diverted and carbon estimates from Section 18.
- **AssistantService.** The language model gets the question and three read tools (search permitted assets, read request status, read indicators). It can propose a request draft but cannot submit it; submission goes through the normal request service after the user confirms.
- **ReportGenerator.** The report service first calculates every figure with the fixed formulas; the language model only writes text around those figures. The gateway checks that every number in the draft equals a standard figure or a change calculated from them (FR-12.4).

5. Human Approval Points

Prepared by: Ahmed AlAbdouli

We put a person wherever a wrong AI answer would be costly or hard to undo: where an asset could be reserved or moved, where a saved record would change, where something could be disposed of, and where a report could be shared. Figure 5-1 shows the gates on the main reuse path, and Table 5-1 lists all of them. At each gate the system records who decided, when, the decision, the reason and what the person saw: the AI result, its confidence and its explanation (FR-7.A4, AI-5).



Grey hexagons with a thick border are human gates: each records who decided, when, the decision, the reason and what the person saw (the AI score, confidence and explanation, FR-7.A4). Check: no path from the AI lane reaches a ReSource record change without passing a gate. No person approves their own request or fills two approval stages (FR-4.4, BR-4).

Figure 5-1: Human approval gates

Table 5-1: Human approval points

ID	Decision	Who decides	What they see	Log entry	Requirements
H-1	Accept, edit or reject each AI classification field	Asset Coordinator of the owning department	Each suggestion, its confidence and reason, conflicting sources	AI output, decision per field, final values	FR-9.6, FR-9.9, FR-1.12
H-2	Choose a match, choose another or reject all	Procurement Team member	Ranked list, score, factor values, confidence, explanation, rule checks	Recommendation decision and reason when not the top match	FR-8.7, AI-3, QUA-12
H-3	Accept a partial or split quantity	Requesting Department Representative	Units available and units requested	Confirmation on the request	FR-8.6
H-4	Release or decline a request	Owning Asset Coordinator (not the requester)	Request, quantity, requester department, the match score if any	Approval record (Release)	FR-4.1, FR-4.4, BR-3
H-5	Approve or reject the released request	Administrator (did not submit or release it)	Request, asset, recheck result, AI score and reasons if from a match	Approval record (Approval)	FR-4.2, FR-4.5, BR-4
H-6	Second approval for high or unknown value	Procurement Team member (third distinct person)	Estimated value and the limit	Approval record (Second approval)	FR-4.3, TBD-2
H-7	Confirm receipt or report a problem; decide the dispute	Receiving DR or AC; Administrator for disputes	Transfer, handling notes, expected condition	Transfer state, condition, decision	FR-4.10, FR-4.12
H-8	Safety review of an end-of-life proposal	Sustainability Officer other than the proposer	Hazard flag, inspection results, AI action advice if any	Safety review, handling party	FR-5.23, FR-5.24, SAF-3
H-9	Approve the end-of-life action	Administrator other than the proposer	Proposal, safety review, AI advice and its confidence	Approval with reason (own reason if advice is Low)	BR-8, SAF-4, SAF-6, FR-10.7
H-10	Approve an AI report before sharing	Finance Officer or Administrator; Sustainability Officer for reports without financial figures	Draft, figure check, assumptions, model version	Original draft, edits and approved version	FR-12.9, FR-12.10, BR-11
H-11	Confirm an assistant action	The Department Representative who asked	Proposed action and final values	Assistant action, confirmed or cancelled	FR-11.3, FR-11.10
H-12	Human review of an AI match or recommendation	Administrator	The AI output, its inputs and the decision taken	Human review record	SEC-10
H-13	Put a new model version or threshold set into use	System Administrator after the evaluation run passes	Evaluation results against the targets in Table 19-1	Model version, approved by	AI-10, QUA-7, QUA-9, QUA-10

The AI cannot do any of the following on its own: grant access, approve a request, reserve units, change custody or location, certify safety, complete an end-of-life action, share a report or change its own thresholds (BR-9, SAF-2). In Figure 5-1, no path from the AI lane reaches a change to a ReSource record without passing through a gate. The system also stops one person from filling two stages of the same approval (FR-4.4, BR-4, SAF-3).

7. Mockups

7.3 Recommendation Review

Prepared by: Ahmed AlAbdouli

The Procurement Team member sees the ranked matches for R-2026-0391. The top match shows its eight factor values, the confidence and a plain-language reason; the second match shows its main limitation. Choosing any other option asks for a reason (FR-8.7).

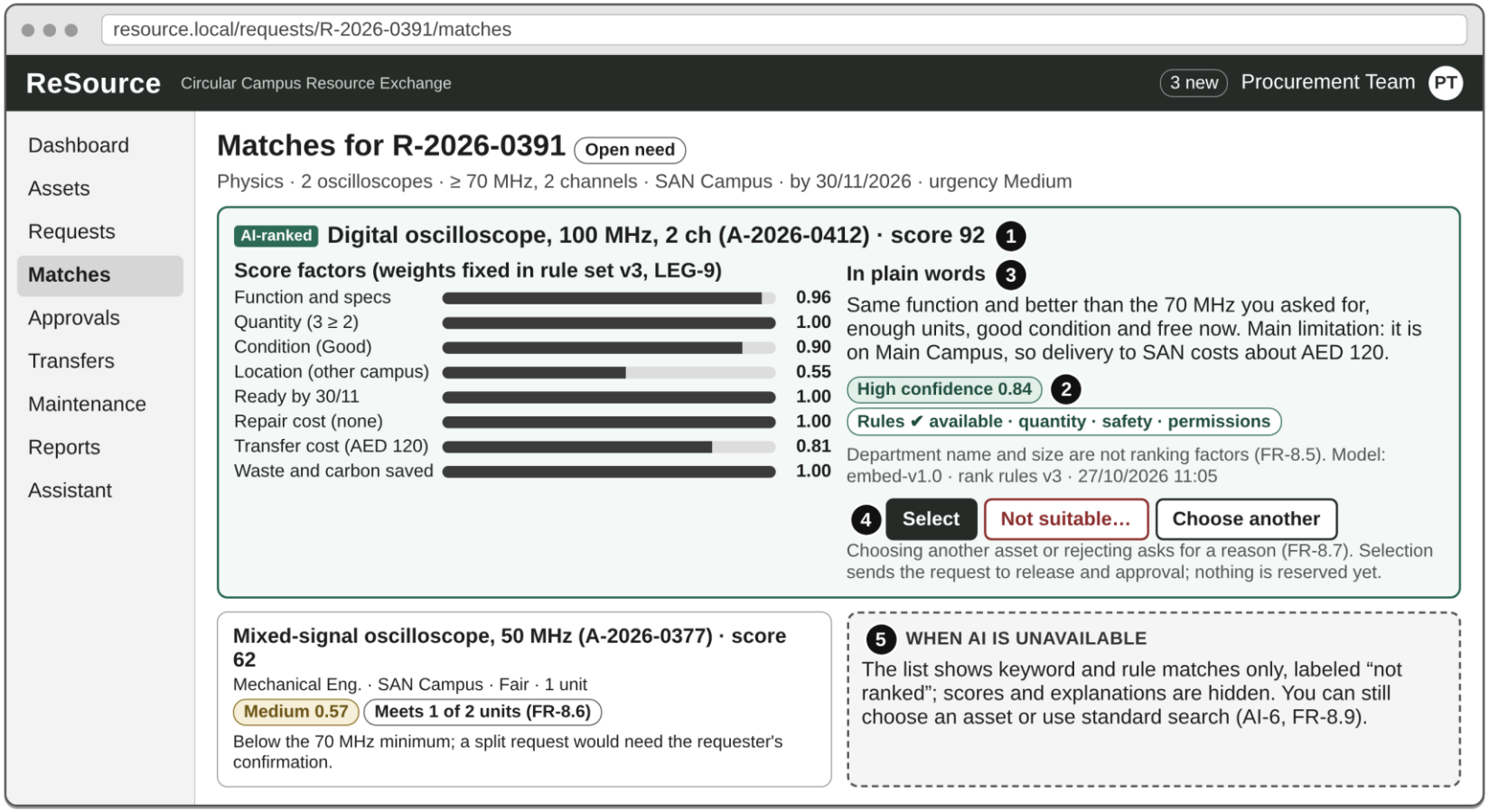


Figure 7-3: Recommendation Review screen

7.8 Conversational Interaction

Prepared by: Ahmed AlAbdouli

The Department Representative asks the assistant in everyday language. The assistant answers from records the user may see, links the listing and prepares a request draft that is submitted only after the user confirms (FR-11.3).

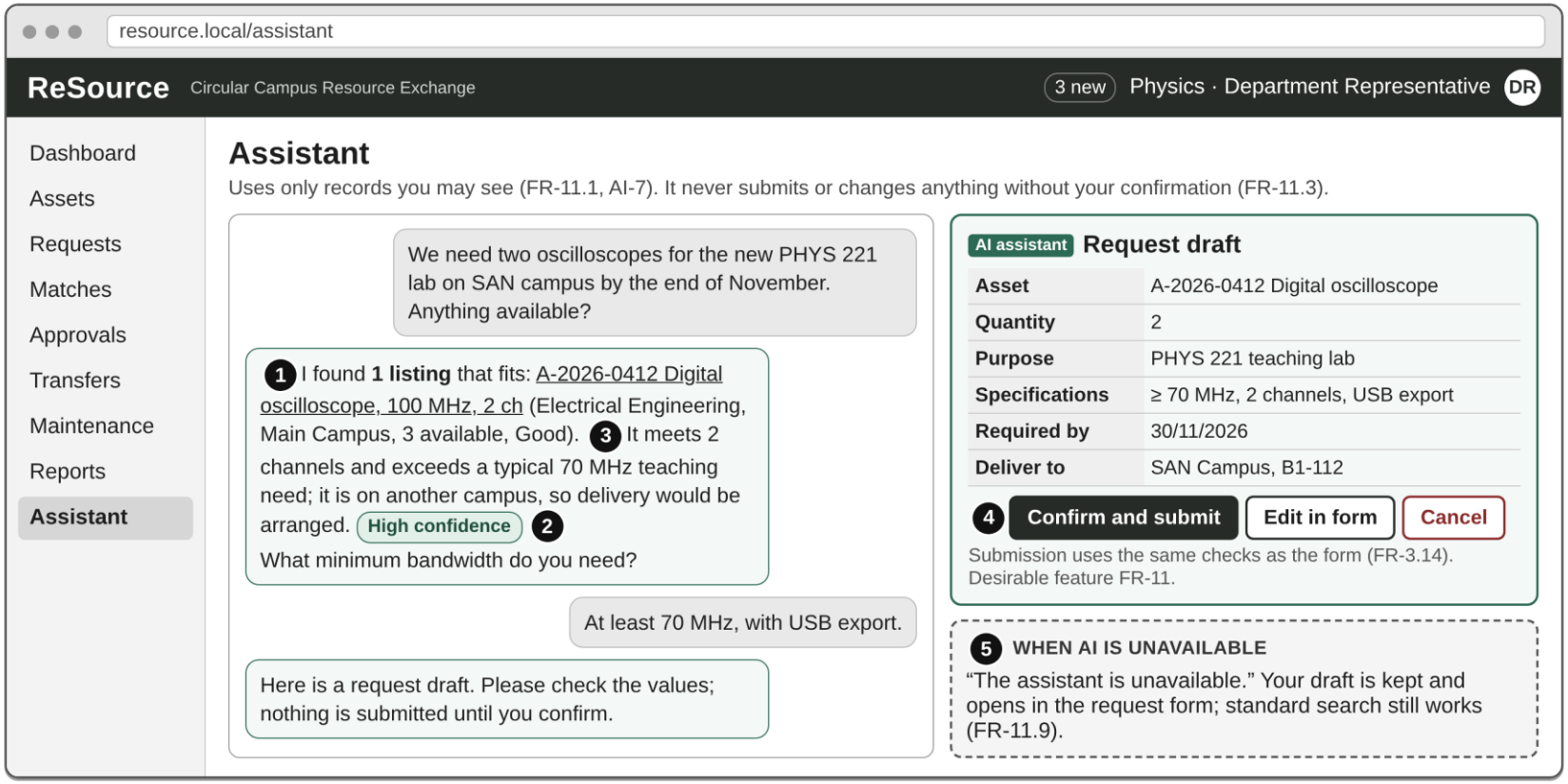


Figure 7-8: Conversational Interaction screen

9. Class Diagrams

9.3 AI Services

Prepared by: Ahmed AlAbdouli

AIGateway is the only class that calls the AI service interfaces and the AI Provider. Every call creates an AIOutput, which always carries a confidence, a level from ConfidencePolicy, an Explanation and the ModelVersion that produced it. `recordDecision` stores the human decision and reason (FR-7.A4). The interfaces are the contracts in Section 16.2.

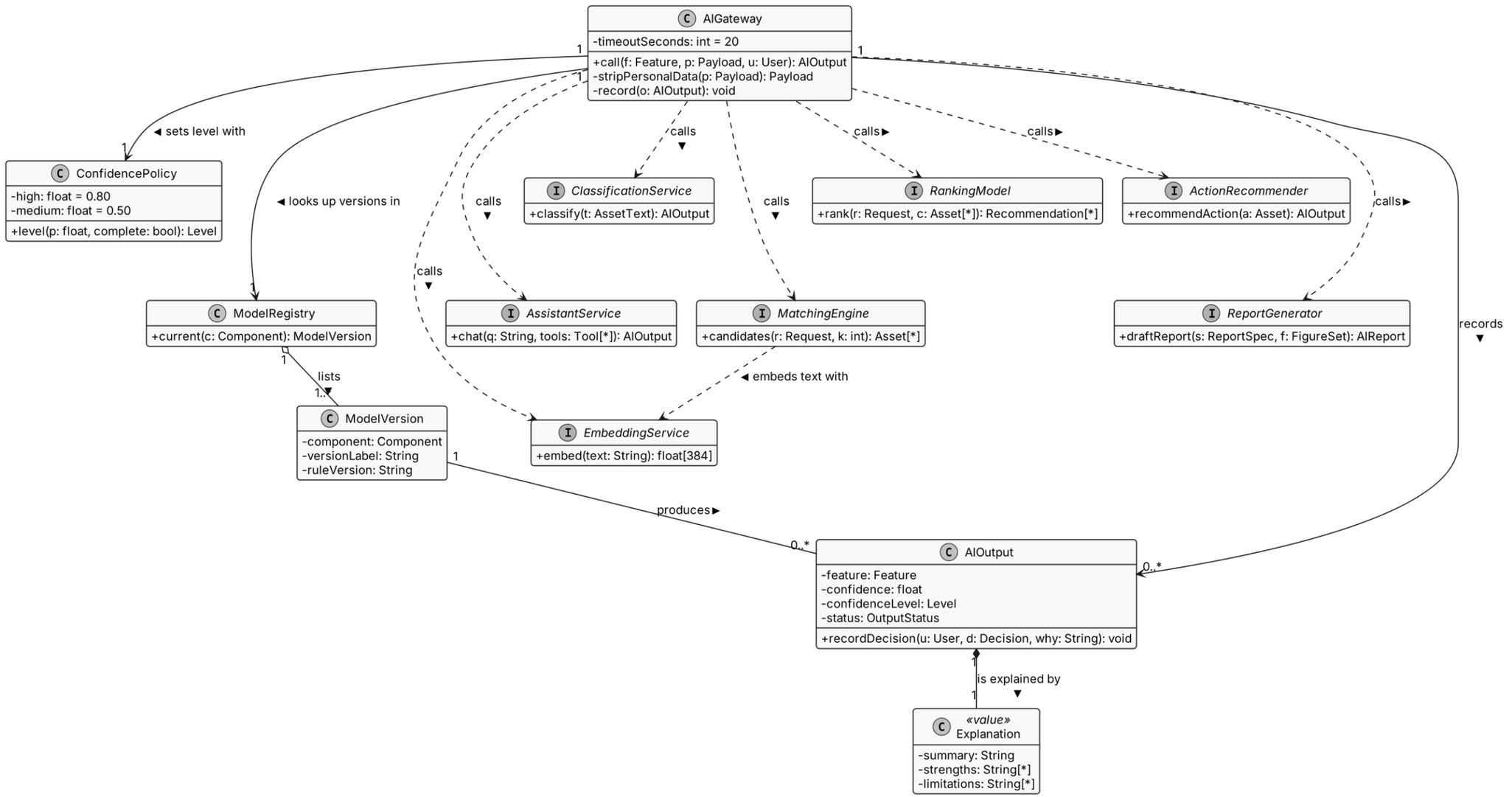


Figure 9-4: Class diagram, AI services

Figure 9-5 shows what the AI services leave behind. One ranking AIOutput ranks up to 20 Recommendations, each linked to the Request, the Asset and the User who reviewed it; an AIReport is drafted by one AIOutput and approved by a User; a HumanReview links the person who asked for it and the Administrator who decides (SEC-10). EvaluationRuns and MonitoringRecords belong to one ModelVersion (QUA-9, QUA-12).

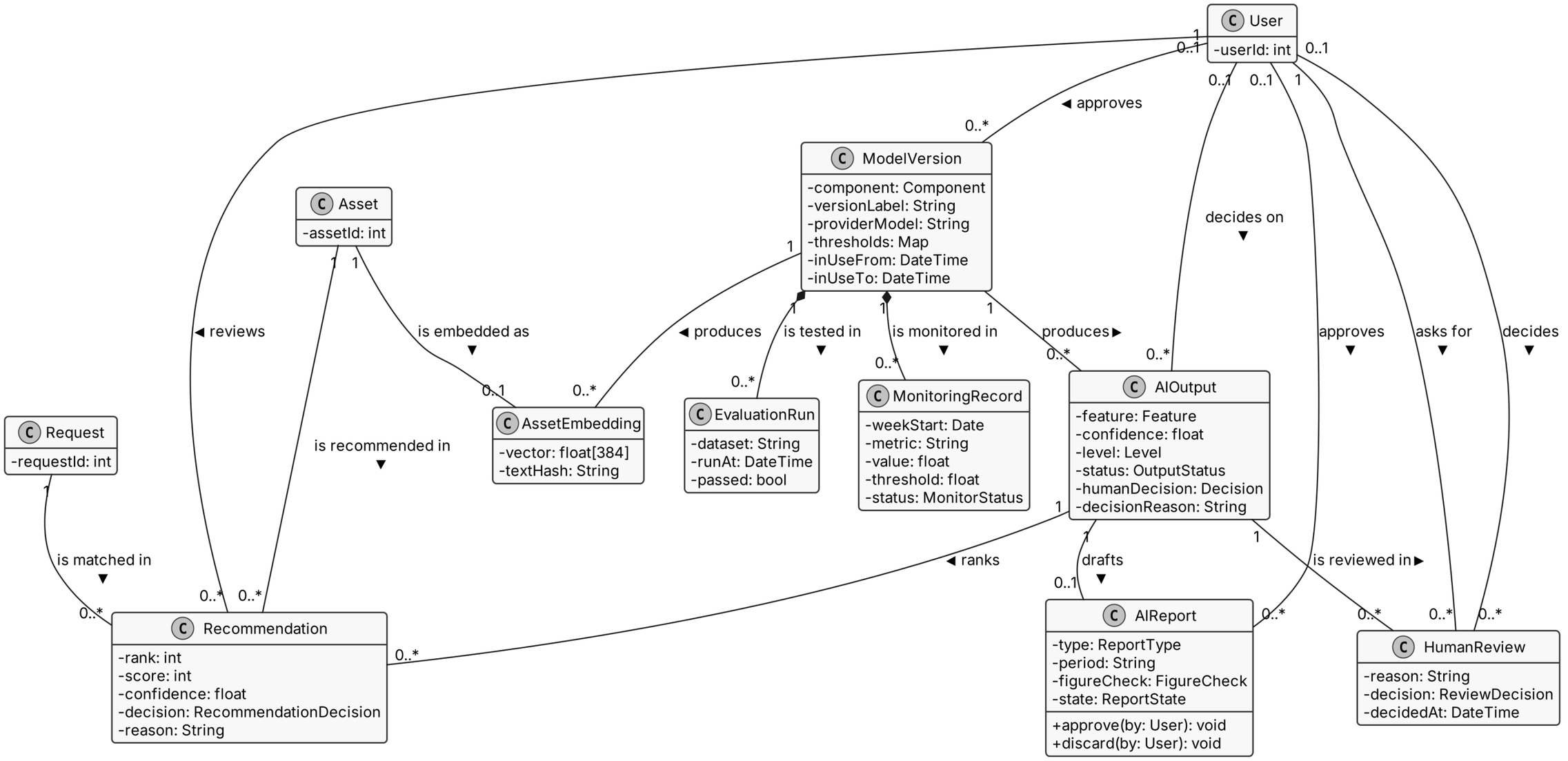


Figure 9-5: Class diagram, AI records

10. Sequence Diagrams

Prepared by: Ahmed AlAbdouli

The sequence diagrams cover all 13 use cases (Table 8-1). Participants are the actors (Phase 3 role names), the screen of Section 7, the modules and AI classes of the component diagram (Figure 13-1), the AI Provider and the MySQL database. Messages to modules are the operations of Table 16-1, and messages to AI services are the operations of Table 16-2. Solid arrows with filled heads are calls, dashed arrows are replies and open-headed arrows are messages to Notifications. "alt" and "opt" frames show the alternative paths, including every fallback. A reject, decline or refusal is always the last branch of its path: after it the diagram shows only the record, the notification and the reply.

Table 10-1: Sequence diagrams

Section	Interaction	Use cases	Figure
10.1	Release and approve a request; second approval	UC-4	10-1, 10-2
10.2	Transfer an asset, including a dispute	UC-5	10-3
10.3	Register, classify and publish an asset	UC-1, UC-10	10-4
10.4	Review AI matches	UC-9	10-5
10.5	Generate an AI report	UC-12	10-6
10.6	Ask the assistant and submit a request	UC-3, UC-11	10-7
10.7	Search and request a listing	UC-2, UC-3	10-8
10.8	Register and activate an account	UC-13, UC-8	10-9
10.9	Record maintenance or repair	UC-6	10-10
10.10	Propose and complete an end-of-life action	UC-7	10-11

10.3 Register, Classify and Publish an Asset (UC-1, UC-10)

Prepared by: Ahmed AlAbdouli

The draft is saved before the AI is called, so a failed or slow AI call never loses the entries (FR-9.8). AIGateway removes personal data and is the only part that sends the prompt to the AI Provider. The suggestion is shown without being accepted; the asset changes only when the Asset Coordinator confirms each field (FR-9.6) and publishes it, and publishing is refused while a required field or a safety check is missing.

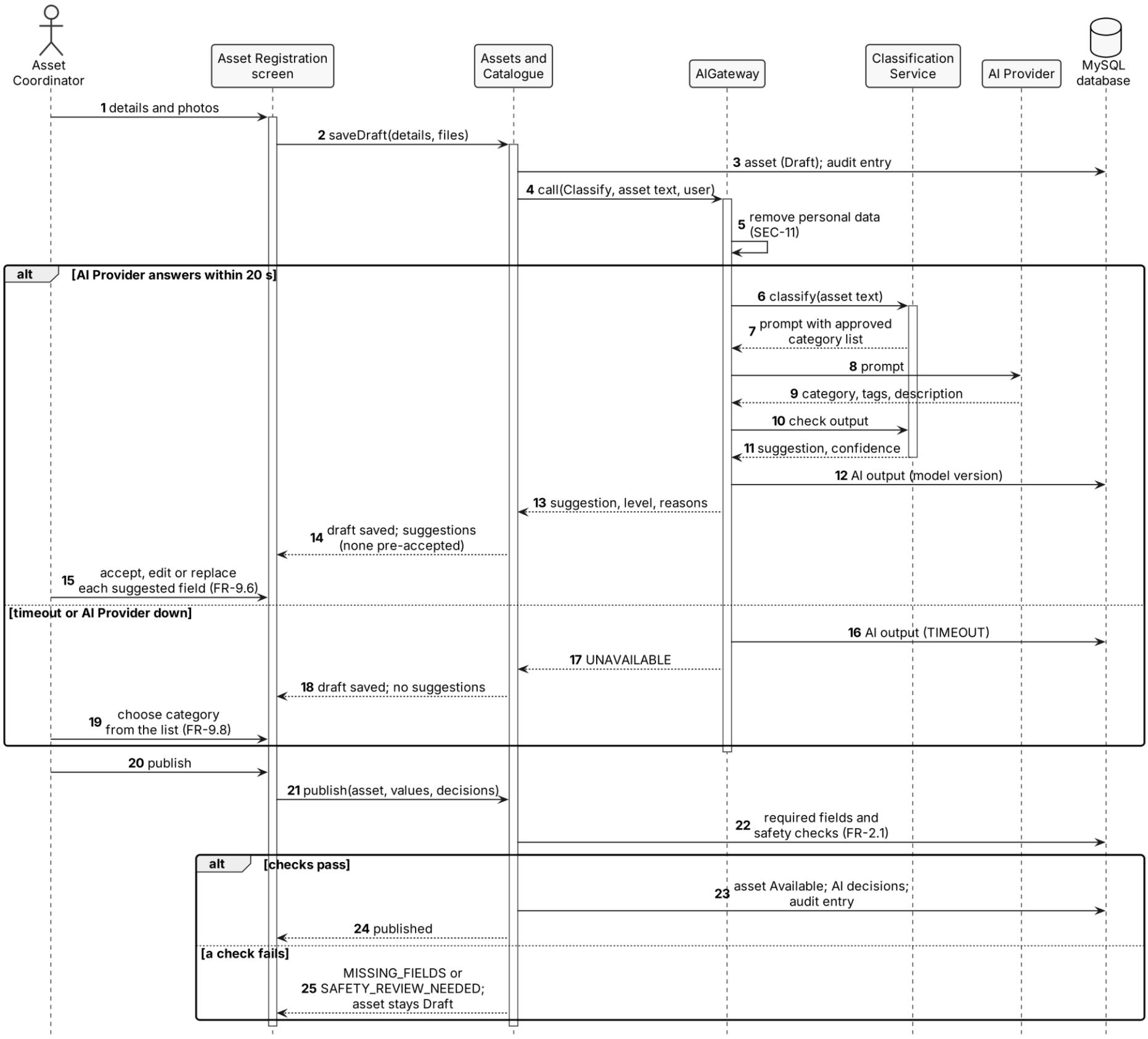


Figure 10-4: Register, classify and publish an asset

10.4 Review AI Matches (UC-9)

Prepared by: Ahmed AlAbdouli

AIGateway gets the 20 most similar eligible assets from MatchingEngine, RankingModel scores them, and each recommendation is stored before the list is shown. If the AI is down or too slow, keyword and rule matches are shown labeled "not ranked" (AI-6). Selecting a match rechecks the asset before the request goes to release (FR-8.8); rejecting all matches closes the request with the reason, and nothing runs after it.

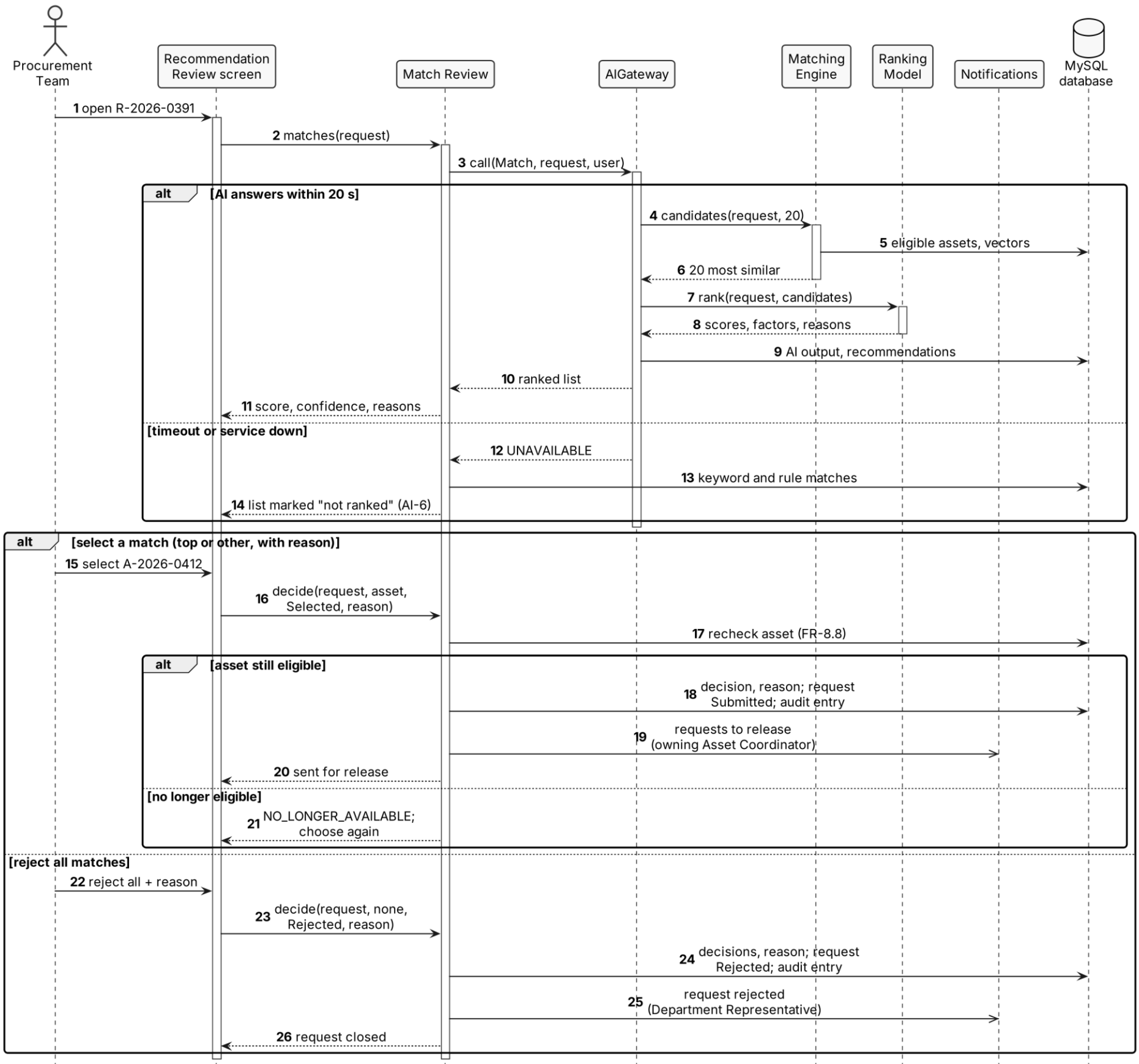


Figure 10-5: Review AI matches

10.6 Ask the Assistant and Submit a Request (UC-3, UC-11)

Prepared by: Ahmed AlAbdouli

The question goes from the screen to AIGateway, the only part that talks to the AI Provider. The assistant reads only records the user may see, through read-only queries filtered by the user's permissions, and the results are passed to the model as data (AI-9). The draft is submitted through the normal request operation, with the same checks as the form, only after the user confirms (FR-3.14, FR-11.3). If the assistant is unavailable, the draft is kept and opens in the request form (FR-11.9).

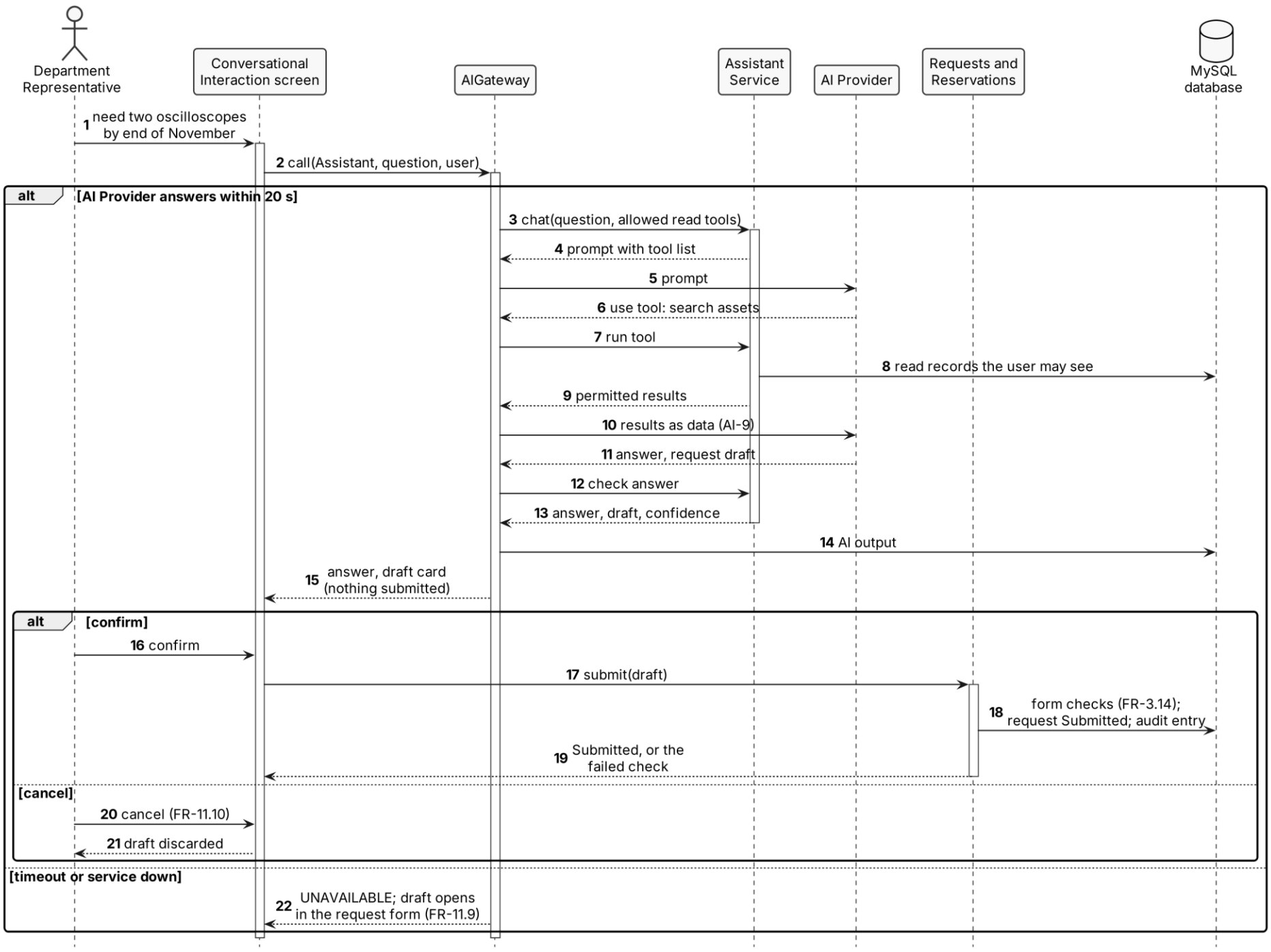


Figure 10-7: Assistant and request draft

14. AI Recommendation Workflow

Prepared by: Ahmed AlAbdouli

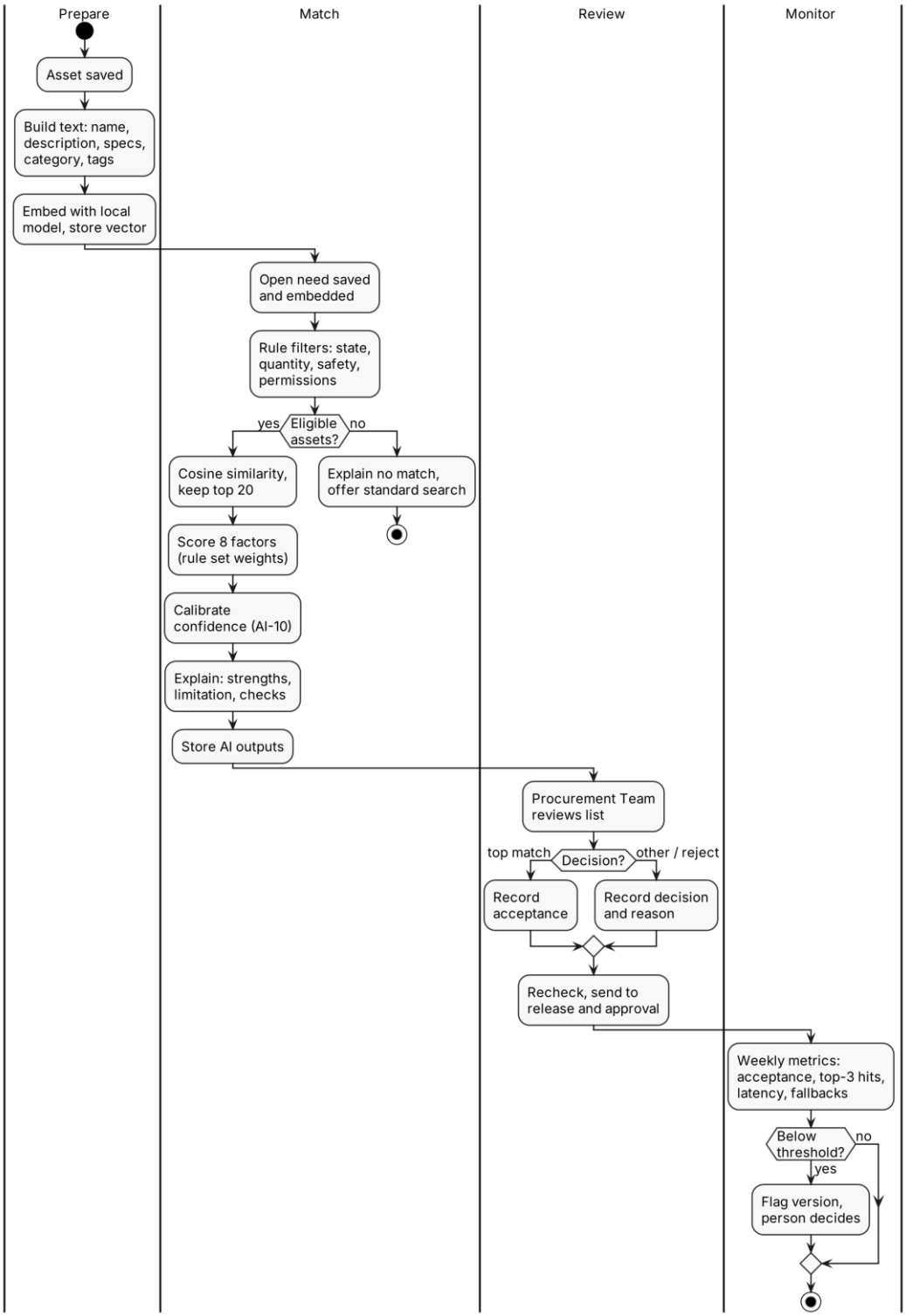


Figure 14-1: AI recommendation workflow

Figure 14-1 shows the full life of a match recommendation, from the asset being saved to the weekly monitoring of the results. It has four stages.

1. **Prepare (pattern B).** When an asset is saved, the worker builds its text (name, description, specifications, category, tags), embeds it and stores the vector with the text hash and model version.
2. **Match.** When a need is saved and embedded, rule filters remove assets that fail state, quantity, safety or permission checks (FR-8.2). Cosine similarity keeps the 20 closest, the ranking scores them on eight factors (Section 17), the confidence is calibrated (AI-10) and an explanation is written. Every result is stored before it is shown (AI-5). If nothing is eligible, the system says so and offers standard search (FR-8.9). If the request vector is not ready yet, the page shows the rule matches marked "not ranked" and refreshes when the job finishes.
3. **Review.** The Procurement Team member decides (H-2). A choice other than the top match, or a rejection, needs a reason (FR-8.7). The chosen asset is rechecked and sent to release and approval (FR-8.8).
4. **Monitor.** Each week the worker calculates the monitoring metrics of Section 19. A metric below its threshold flags the model version; a person decides what to do (H-13). Nothing is retrained automatically (FR-9.9).

16. Interface Contracts

16.2 AI Service Contracts

Prepared by: Ahmed AlAbdouli

Modules, and the assistant screen, call AIGateway with a common envelope. The envelope stays inside ReSource; for the three features that use the AI Provider, only the fields in the "payload" go to the provider, after redaction. Table 16-2 lists the seven contracts, one for each operation of the interfaces in Figure 9-4. None of them contains a name, university ID or e-mail field (DR-4).

Table 16-2: AI service contracts

Operation	Request payload	Response	Leaves the server?
classify	asset text (name, description, specifications), photo references, approved category list	category, subcategory, tags, description, missing fields, confidence, reasons, conflicts	Text and photos to the AI Provider
embed	text (up to 256 word pieces)	384-number vector, model version	No
candidates	request_id, number of candidates (20)	eligible asset_ids with cosine similarity	No
rank	request_id, top-20 candidate asset_ids, urgency	ranked list with score, factors, confidence, explanation, rule checks	No
recommendAction	asset_id, condition, safety status, maintenance cost, open requests count	action options with score, confidence, estimates, explanation	No
chat	question, conversation so far, allowed read tools	answer, links, optional request draft, confidence	Question and tool results to the AI Provider
draftReport	report type, period, scope, standard figures	draft text, figure check, assumptions, action list with confidence	Figures and template to the AI Provider

Example request from Match Review to AIGateway for a ranking. AIGateway checks acting_user itself and never sends it to an AI service:

```
{
  "feature": "rank",
  "call_id": "AIQ-2026-31877",
  "acting_user": 1047,
  "time_limit_s": 20,
  "payload": {
    "request_id": "R-2026-0391",
    "urgency": "Medium",
    "candidates": ["A-2026-0412", "A-2026-0377", "..."]
  }
}
```

Example response, showing the first result only (Table 16-3 lists the other status values):

```
{
  "call_id": "AIQ-2026-31877",
  "status": "OK",
  "model_version": "embed-v1.0 / rank rules v3",
  "latency_ms": 640,
  "results": [{
    "asset_id": "A-2026-0412", "rank": 1, "score": 92,
    "factors": {"function": 0.96, "quantity": 1.00, "condition": 0.90,
                "location": 0.55, "ready_by": 1.00, "repair_cost": 1.00,
                "transfer_cost": 0.81, "waste_carbon": 1.00},
    "confidence": 0.84, "confidence_level": "High",
    "explanation": {
      "summary": "Same function and better than the 70 MHz asked for.",
      "strengths": ["function and specs", "quantity", "ready now"],
      "limitation": "On Main Campus: delivery to SAN costs about AED 120.",
      "rule_checks": ["available", "quantity", "safety", "permissions"]
    }
  }],
  "weak_matches": false,
  "ai_output_id": "AI-2026-08812"
}
```

Table 16-3: AI error statuses

Status	Meaning	Caller action
TIMEOUT	No usable result within 20 s	Show the fallback of Table 4-1; record the failed output
UNAVAILABLE	Service marked down after repeated failures	Show the fallback at once
INVALID_OUTPUT	Result failed validation (category not on the list, figure mismatch, bad format)	Treat as failed; show the fallback
BLOCKED	Request refused by an access or content check (AI-7, AI-9)	Tell the user what is not allowed; offer the standard function

17. Recommendation Score and Explanation Design

Prepared by: Ahmed AlAbdouli

17.1 Score

Prepared by: Ahmed AlAbdouli

Each eligible candidate gets a compatibility score from 0 to 100 (FR-8.4), the weighted sum of eight factor values, each between 0 and 1:

```
Score = round( 100 x (w1 F1 + w2 F2 + ... + w8 F8) ),  with w1 + ... + w8 = 1
```

The factors cover every criterion listed in FR-8.3 and FR-8.5. The weights belong to a versioned rule set (rule set v3), are visible to Administrators and can be changed only by creating a new version (LEG-9, QUA-7). The requesting department's name, ID and size are not inputs (LEG-9, QUA-13).

Table 17-1: Score factors (rule set v3)

Factor	Weight	Value between 0 and 1	Criterion
F1 Function and specifications	0.35	clamp((cosine - 0.30) / 0.55, 0, 1) x 0.5 to the power of the number of stated specifications not met	Category, purpose, specifications
F2 Quantity	0.15	min(available, requested) / requested	Quantity (FR-8.6)
F3 Condition	0.10	New 1.0, Good 0.9, Fair 0.6, Poor 0.3; at most 0.3 if below the requested minimum	Current and requested condition
F4 Location	0.10	Same building 1.0, same campus 0.85, other campus 0.55	Location
F5 Ready by the date	0.10	1.0 if available by the required date, 0.5 if up to 2 days late, else 0	Required date, urgency
F6 Repair cost	0.08	1 - (expected repair cost / replacement price), at least 0	Repair needs
F7 Transfer cost	0.07	1 - (transfer cost / (0.15 x replacement price)), at least 0	Transfer cost
F8 Waste and carbon saved	0.05	carbon avoided by this candidate / largest carbon avoided among the candidates	Waste and carbon reduction

Urgency changes the weight of F5: High urgency doubles it and Low urgency halves it, and all weights are then rescaled to add up to 1. If no candidate scores 60 or more, the list is shown with a "weak matches" warning and no top pick. A candidate that meets only part of the quantity is shown with that fact (FR-8.6).

17.2 Confidence

Prepared by: Ahmed AlAbdouli

The score ranks candidates; the confidence says how likely it is that the Procurement Team will find the candidate suitable. It is calculated with Platt scaling [14], a logistic curve fitted on the labeled evaluation set of QUA-10:

```
confidence = 1 / (1 + exp(-(a x S + b))),   S = score / 100
example fit: a = 4.60, b = -2.57 (refitted for each model version)
```

The level then follows Table 4-2. A candidate with missing required information (for example no condition or no location) is Low whatever its score (AI-10).

17.3 Explanation Structure

Prepared by: Ahmed AlAbdouli

Every recommendation has the same explanation structure, stored in AI_OUTPUT and shown on the screen (AI-2, QUA-14):

Table 17-2: Explanation fields

Field	How it is produced	Example (A-2026-0412)
summary	One sentence from a template using the two strongest factors	Same function and better than the 70 MHz you asked for, enough units.
strengths	The three factors with the largest weighted values (w x F)	Function and specs; quantity; ready now
limitation	The factor with the largest weighted gap, w x (1 - F)	On Main Campus: delivery to SAN costs about AED 120
rule_checks	Checks passed before scoring	Available, quantity, safety, permissions
confidence, level	Section 17.2	0.84, High
versions	Embedding model and rule set	embed-v1.0, rank rules v3

17.4 Worked Example

Prepared by: Ahmed AlAbdouli

For request R-2026-0391 (2 oscilloscopes, at least 70 MHz, minimum condition Good, SAN Campus, urgency Medium), Table 17-3 scores the two candidates shown in mockups 7.2 and 7.3.

Table 17-3: Worked scoring example

Factor (weight)	A-2026-0412: 100 MHz, Good, 3 units, Main Campus	A-2026-0377: 50 MHz, Fair, 1 unit, SAN Campus
F1 (0.35)	cosine 0.83, all specs met: 0.96	cosine 0.795, 1 spec not met: 0.90 x 0.5 = 0.45
F2 (0.15)	min(3, 2) / 2 = 1.00	1 / 2 = 0.50
F3 (0.10)	Good: 0.90	Fair below the Good minimum: 0.30
F4 (0.10)	Other campus: 0.55	Same campus: 0.85
F5 (0.10)	Ready now: 1.00	Ready now: 1.00
F6 (0.08)	No repair: 1.00	No repair: 1.00
F7 (0.07)	1 - 120 / (0.15 x 4,200) = 0.81	1 - 40 / (0.15 x 3,000) = 0.91
F8 (0.05)	119.2 kg CO2e, the largest: 1.00	68.8 / 119.2 = 0.58
Score	100 x 0.919 = 92	100 x 0.620 = 62
Confidence	0.84, High	0.57, Medium

Carbon avoided uses Section 18: 2 units x 2.6 kg x 22.92 kg CO2e per kg = 119.2 for A-2026-0412 and 1 unit x 3.0 kg x 22.92 = 68.8 for A-2026-0377. Transfer costs (AED 120 and AED 40) and the AED 3,000 replacement price of A-2026-0377 are sample values.

19. Model-Monitoring Records

Prepared by: Ahmed AlAbdouli

Two kinds of record show whether the AI keeps meeting its targets. An EVALUATION_RUN tests a model version on a fixed labeled data set before it is put into use and after any change. A MONITORING_RECORD stores one weekly metric from real use. Both point to the MODEL_VERSION they measure.

Table 19-1: Monitoring metrics and thresholds

Metric	How it is calculated	Source	Target or threshold	Requirements
Top-3 hit rate	Test requests with a known match whose match is in the first three	Evaluation run	At least 80%	QUA-10
Category accuracy	Test assets whose main category is suggested correctly	Evaluation run	At least 80%	QUA-9
First action acceptable	Test scenarios where the first recommended action is acceptable	Evaluation run	At least 80%	QUA-11
Department-name test	Rankings unchanged when only the department name or ID changes	Evaluation run	100% unchanged	QUA-13, LEG-9
Assistant correctness	Correct answers and correct refusals on the test questions	Evaluation run	At least 80% (TBD-12)	QUA-15
Calibration error	Average gap between confidence and actual acceptance, in 10 bins	Evaluation run	0.10 or less	AI-10
Match acceptance rate	Top matches accepted / match decisions, also by confidence level	Weekly	Amber below 60%, red below 40%	QUA-12
Override reasons	Count of each reason for choosing another match or rejecting	Weekly	Reviewed when one reason exceeds 30%	FR-8.7
Classification acceptance	Suggested fields accepted without edit / fields suggested	Weekly	Amber below 60%	FR-9.9
Response time	AI calls with a usable result within 15 s / all calls	Weekly	At least 90%	PER-4
Fallback rate	Calls ending in TIMEOUT, UNAVAILABLE or INVALID_OUTPUT	Weekly	Amber above 5%, red above 10%	AI-6
Report figure failures	AI report drafts rejected by the figure check	Weekly	Any failure is reviewed	FR-12.4

Table 19-2: Example monitoring records

week_start	model_version	metric	value	threshold	status
19/10/2026	rank rules v3	match_acceptance_rate	0.71	0.60	Green
19/10/2026	rank rules v3	response_time_within_15s	0.97	0.90	Green
19/10/2026	classify-v1.2	classification_acceptance	0.58	0.60	Amber
19/10/2026	classify-v1.2	fallback_rate	0.03	0.05	Green

An amber result notifies the System Administrator and the Administrator, who review the overrides and reasons. A red result, or a failed evaluation run, stops the version for new suggestions until a person decides to restore the previous version, publish a new rule set or thresholds, or accept the result with a reason (H-13). The AI is never retrained automatically (FR-9.9).

AI Contributions

Ahmed AlAbdouli

Prepared by: Ahmed AlAbdouli

I used an AI assistant (ChatGPT astra6, OpenAI) to help draw the AI class diagrams (Section 9.3), the sequence diagrams in Sections 10.3, 10.4 and 10.6, the recommendation workflow (Section 14) and the screen mockups in Sections 7.3 and 7.8 from my design, and to check the AI requirements, time limits and the worked example against the Phase 3 report. I wrote the section content and reviewed every figure.

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